Introduction to medical research: Essential skills

Module 1. Research planning: before you start your research

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NDORMS
NUFFIELD DEPARTMENT OF ORTHOPAEDICS, RHEUMATOLOGY AND MUSCULOSKELETAL SCIENCES

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Objectives

- By the end of today you should have:
  - Clearer idea of what medical research involves
  - What are the key ethical and governance issues
  - How to turn research idea into a specific research question
  - How to systematically collect and synthesise literature to support further research project design and planning

  - Course is build around **competencies** specified in the **Academic Compendium** of the UK Foundation Programme

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Importance of planning

- Scientific aspects of my research project
- 'Administrative' aspects (approvals – finance, R&D, ethics, ..)
- Constraints, logistics (time, money, personnel, delays,..)

Typical research process: key steps

Conception

Design

Execution

Analysis

Publication and reporting

Key elements of good research

Conception

Design

Execution

Analysis

Publication and reporting

- Turning idea into a research question (clear research aim, rationale, hypothesis)
- Treatment – clinical equipoise
- Consult study with ‘users’ – patients’ input (questions, design - outcomes, process, ..)
- Systematic review of existing evidence
- Consider research participants: practicalities; wellbeing; courtesy; respect

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Key elements of good research

- Conception
- **Design**
- Execution
- Analysis
- Publication and reporting

- CONSULT with a STATISTICIAN
- Develop appropriate design to answer question
- Adequate sample size
- Adequate steps to reduce bias
- Carefully balance risk and benefits; inclusion and exclusion criteria
- Develop a protocol
- Ethical approval – informed consent, PIS, questionnaires, ..

Key elements of good research

- Conception
- Design
- **Execution**
- Analysis
- Publication and reporting

- Participants' recruitment and **consent** form – trained personnel (voluntary, avoid coercion, right to withdraw, fully informed, ..)
- Good clinical practice conduct, research integrity principles (data collection, data protection and confidentiality)

Key elements of good research

- Conception
- Design
- Execution
- **Analysis**
- Publication and reporting

- Data analysis should be hypothesis driven (statistical analysis plan)
- A statistician should be employed from the study conception
- Avoid any data fabrication / falsification

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Key elements of good research

- Conception
- Design
- Execution
- Analysis
- Publication and reporting

- Research only has **value** if
  - Study methods have validity
  - Research findings are published in a usable form
- The goal should be **transparency** and honest account
  - Should not mislead
  - Should allow replication (in principle)
- **Impact** - research paper needs to be usable, 'fit for purpose'
  - we need to think beyond our own work
- **Reporting guidelines:** www.equator-network.org

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Types of clinical research

- Observational studies
  - Case reports
  - Surveys
  - Cohort studies
  - Cross-sectional studies
  - Case-control studies
- Experimental studies
  - Randomised trials
  - Non-randomised studies
- Qualitative research
- Research synthesis (systematic reviews)

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Types of studies by research questions / focus

- Treatment evaluations - RCTs
- Disease aetiology, harms, .. - Observational studies

- Prognostic studies
- Diagnostic studies
- Experiences, views, .. – Qualitative studies
- Quality improvement studies
- Economic evaluations
- Patients’ involvement in research

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Medical research: where does my project fit in?

Laboratory research
Animal pre-clinical research

Phase I, II trials

Phase III trials
Observational research

Qualitative research, Diagnostic & prognostic; economic analyses, surveys

Research synthesis / Systematic reviews

Clinical guidelines

Phase IV / pragmatic trials
Outcomes research

Different types of research designs

- Some research designs are more suitable for answering a given research question than others - important to choose an appropriate research design!
  - E.g. RCT – treatment evaluation

- Different advantages but also limitations

- Bring different ethical issues
  - E.g. – accepting concept of randomisation

Principles guiding medical research

- Ethical guidelines are important in clinical research
  - They safeguard participants' health, safety and privacy
  - They help in building public trust in medical research (unethical practices lead to wasting time and money and increased regulation – burden)
- Ethics –very broad term

Research project example 1

- Barriers and opportunities for enhancing recruitment and retention of Asian patients in clinical trials conducted in London: a survey of clinical staff, patients and volunteers
- Discuss potential
  – Ethical issues
  – Logistics issues

Research project example 2

- Researchers at a UK university wish to undertake a whole genome sequencing study of individuals from a population who live on a remote island off the British coast. The researchers have a particular interest in looking at the occurrence of neurological disorders in this population. The researchers have been made aware that there is a pre-collected DNA sample set available which the sample custodian would be happy to share with them for this study.

- Discuss potential
  - Ethical issues , logistics issues

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Declaration of Helsinki: Key ethical principles

- General principles
- Risks, burdens and benefits
- Vulnerable groups and individuals
- Scientific requirements and research protocols
- Research ethics committees
- Privacy and confidentiality
- Informed consent
- Use of placebo
- Post-trial provisions
- Research registration and publication and dissemination of results
- Unproven interventions in clinical practice

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Research governance

- Broad range of regulations, principles and standards of good practice that exist to achieve, and continuously improve, research quality in the UK and worldwide

- Key documents:
  - Research Governance Framework
  - Good Clinical Practice

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Research governance (2)

- Key documents:
  - EU Directives (Clinical Trial Directive; Good Clinical Practice Directive)
  - Human Tissue Act 2004
  - Data Protection Act 1998
  - Mental Capacity Act 2005

https://www.admin.ox.ac.uk/researchsupport/ctrg/governance/

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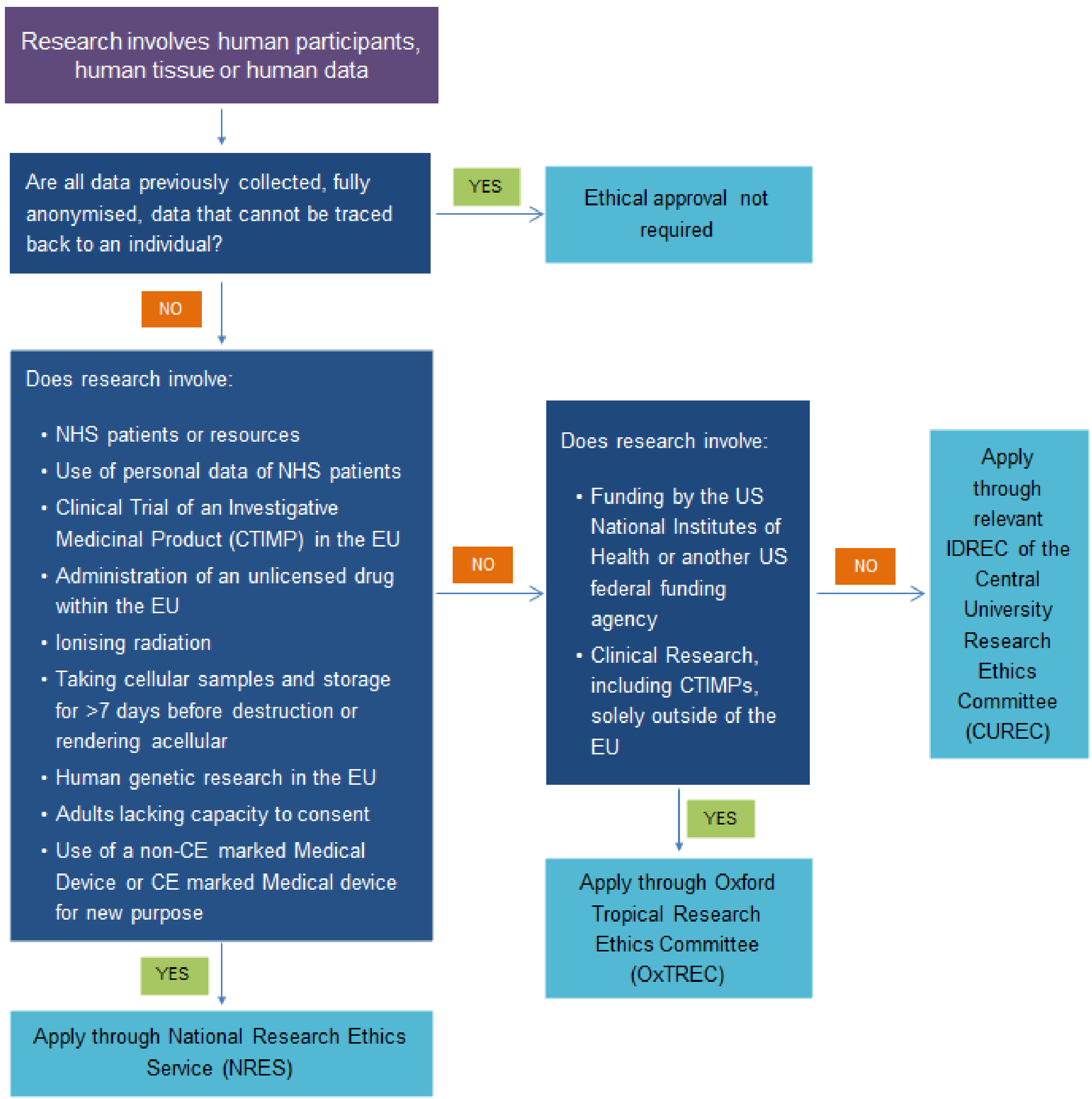
Do I need an ethical approval?

Flowchart



http://www.admin.ox.ac.uk/researchsupport/ctrg/sponsorship/ethics/

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UAS Home > Research Support > Clinical Trials and Research Governance >

- About research support
- Contacts
- News
- Consultations
- Research Committee
- Training and professional development
- Find funding
- Major sponsors
- Applying for funding
- Costing and pricing
- X5: Costing and pricing system
- Research Facilities and Equipment
- Research contracts
- Consulting activity
- Clinical Trials and Research Governance
  - About us
  - Research Governance
  - Research Classification and Procedures
  - Sponsorship and Ethics Applications
  - Resources
  - Training
- Intellectual Property
- Knowledge Exchange and Impact

Clinical Trials and Research Governance

The Clinical Trials and Research Governance (CTRG) team supports University Clinical Researchers. CTRG has a role in promoting and supporting high quality research within the University, and ensuring the University meets its requirements as Sponsor and host institution.

About CTRG

CTRG

Research Governance

Research Classification and Procedures

Sponsorship and Ethics Applications

Resources

Training

FAQs

- What to consider when planning research
- CUREC or NHS Ethical Review?
- What documents do I need to prepare?
- What is a research protocol, and why do I need one?
- Is MHRA approval required?
- Is NHS permission required?

CTRG
Clinical Trials and Research Governance

Contacts

- Contact the CTRG team

Find us

- Directions to CTRG team, Joint Research Office (836kb)

Map

- Click here for a Google Map

Documents

- Glossary of acronyms used on this site (141kb)
- Guidance for Researchers Seeking Sponsorship (743kb)
- Template PIS and Consent Form (149kb)
- IRAS Guidance (744kb)

Related links

- Safety Reporting (Serious Adverse Events)
- Clinical Trial Units
- Central University Research

http://www.admin.ox.ac.uk/researchsupport/ctrg/

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RESEARCH SUPPORT
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Research Services

Find funding

Major sponsors, including EC

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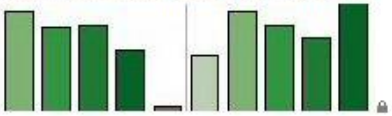
Clinical Trials & Research Studies

CTRG

Research integrity and ethics

Business Intelligence (BI) Reports



Research Contracts

Back to top

Key contacts

- Research Services
- Research Accounts
- Isis Innovation Limited
- Research Facilitators

X5 Research costing and pricing system

Oxford Gateway to Europe

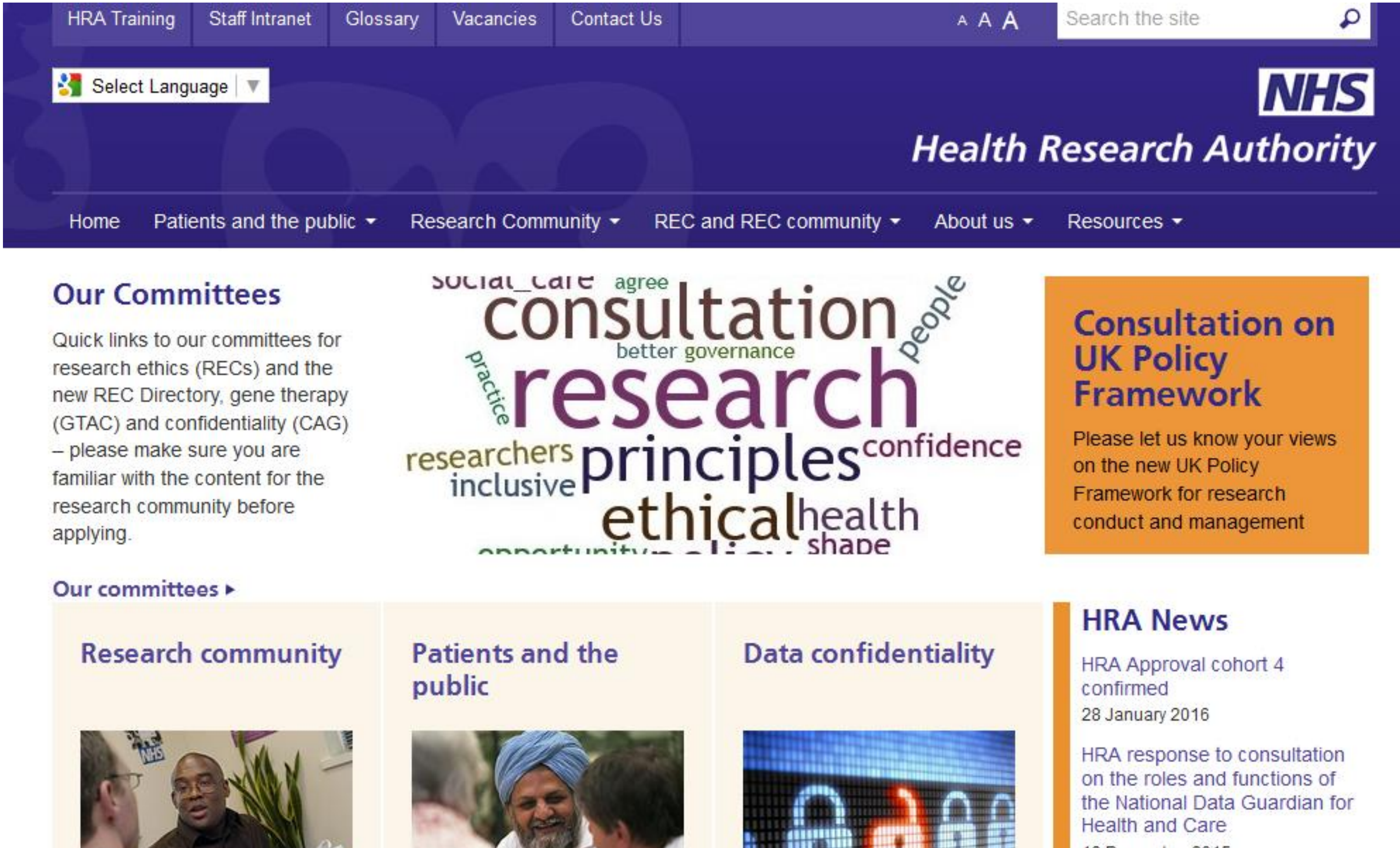
oao open access oxford
Expanding the possibilities

RESEARCH DATA OXFORD

ORCID

Other useful resources

HRA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About us
Resources
Our Committees
Quick links to our committees for research ethics (RECs) and the new REC Directory, gene therapy (GTAC) and confidentiality (CAG) – please make sure you are familiar with the content for the research community before applying.
agree
consultation
people
better governance
practice
research
researchers
principles
confidence
inclusive
ethical
health
shape
Consultation on UK Policy Framework
Please let us know your views on the new UK Policy Framework for research conduct and management
Our committees ▸
Research community
Patients and the public
Data confidentiality
NHS
HRA News
HRA Approval cohort 4 confirmed
28 January 2016
HRA response to consultation on the roles and functions of the National Data Guardian for Health and Care

Resources

Resources pages provide additional detail on important topics. This includes links to external sites, reference documents and explanatory text.

Many pages also provide links to other areas of the site, where related information can be found.

We have created these to give you extra detail on key themes, and many of them can be reached from more than one page in the site.

You can find information in this section using the alphabetical list below, the categories in the menu on the left or using the search box.

List of Resources

- Adding new sites to a study
- Administration of Radioactive Substances
- Adults Unable to Consent for Themselves
- Amendments
- Applying for Approvals: Template Documents
- Care After Research
- Chief Investigator
- Clinical Research Networks
- Clinical Trials of Investigational Medicinal Products (CTIMPs)
- Confidentiality Advisory Group (CAG) – Annual Review Template
- Confidentiality Advisory Group (CAG) – Meeting Dates
- Confidentiality Advisory Group (CAG) – Pre-application Decision Tool
- Confidentiality Advisory Group (CAG) – Standard Conditions of Support
- Confidentiality Advisory Group (CAG) – Application Advice
- Confidentiality Advisory Group (CAG) – Proportionate Review
- Confidentiality Advisory Group (CAG) – Section 251 form for non-research applications
- Confidentiality, Privacy and Data Protection
- Consent and Participant Information
- End of Study Notification – Clinical Trials of Investigational Medicinal Products (CTIMPs): EudraCT form
- End of Study Notification – Studies other than Clinical Trials of Investigational Medicinal Products
- Ethical Considerations in Research
- Ethical Review for Private and Voluntary Health Care

Key aspects of health research ethics: simple practical checklists

General Considerations

Specific Considerations

Participant Information Sheets

Consent Forms

Withdrawn Consent

Renumeration

Mental Capacity

Research with Children

Internet Mediated Research

Internet Mediated Research – Top Tips

Research Design Service
South Central

NHS
National Institute for
Health Research

Home | About us | Funding | Training | Public Patient Involvement | Planning a study | Resources | Latest news | FAQs | Contact | Internal

"We provide free advice on research design to researchers in the South Central region who are developing proposals for national, peer-reviewed funding competitions for applied health or social care research

How we can help

If you have a research question you would like to turn into a fully developed funding proposal, we can help. Our experienced Research Advisors have expertise in key aspects of preparing grant applications, including:

- Design
- Methodology
- Identifying an appropriate funder
- Involving patients and public
- **Costing your research project**

Who we can help

Read on and find out if you are eligible for our support.

Find out more about who the RDS can help and how it supports researchers in the RDS leaflet

REQUEST SUPPORT

What you can expect from us

The RDS Charter provides information about the services we offer.

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Planning a study

Information, tips and pointers to other resources to help your research team plan an effective study.

Specialist advice on specific aspects of a study

Follow the links on the right.

Getting started

For a general overview of what's involved when planning a study, get started with our Information Pack for Health and Social Care Researchers

This highlights some of the key issues to consider when making a research grant application, including:

- Refining your research idea
- Examining the current literature
- Getting your methodology right
- The usual structure of a grant application, and what a reviewer would expect to see in each section
- Logistical issues to consider, including time and money
- Types of costs typically requested
- Ethics and governance procedures required of all NHS research
- What Patient and Public Involvement is, and how to do it well
- Other organisations which may be useful to you.

This list is not definitive; your Research Advisor can provide advice specifically tailored to your application.

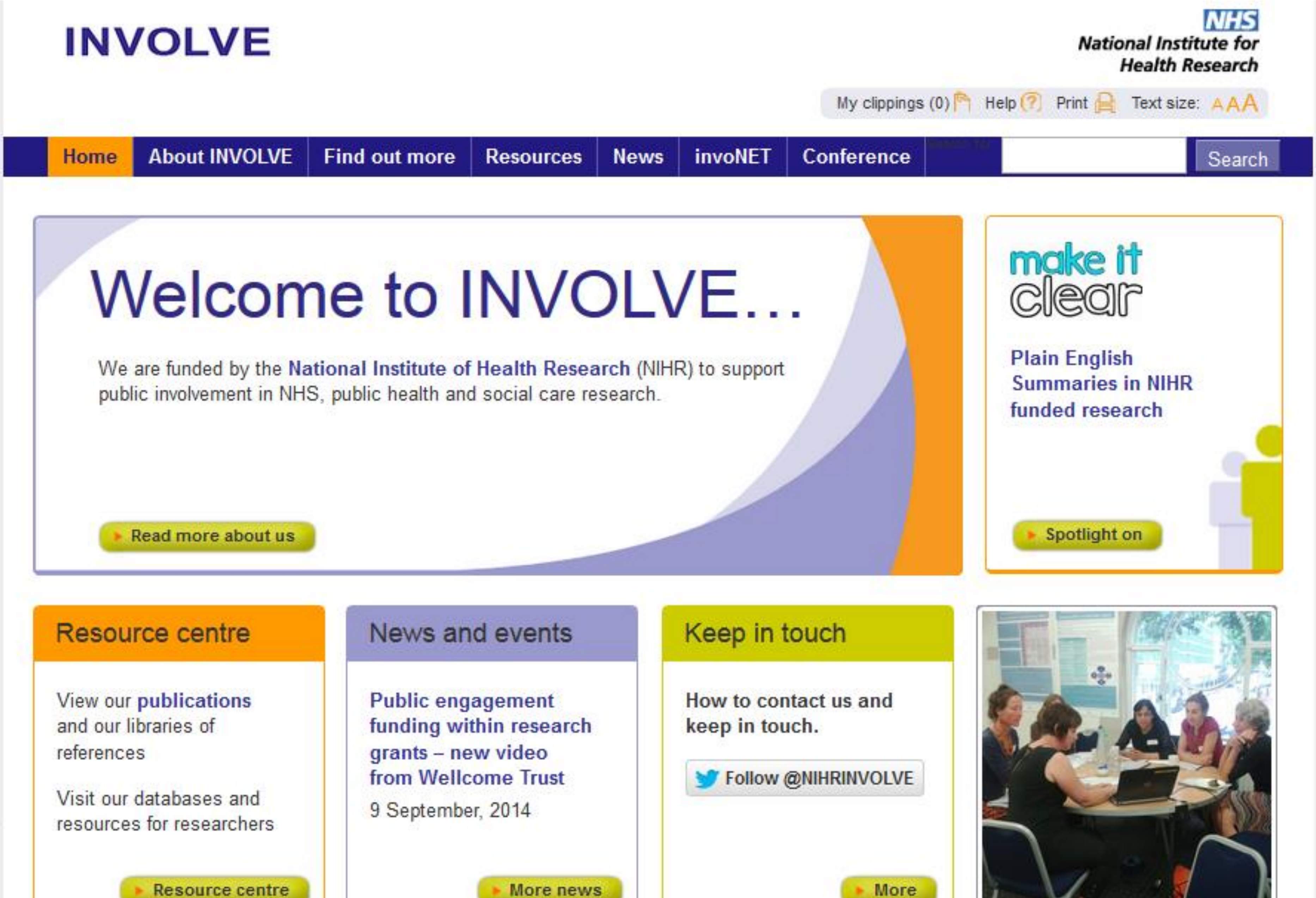
Download the full Information Pack for Health and Social Care Researchers

Share the knowledge

If you discover a useful resource not mentioned here, please drop us a line at rds.sc@nihr.ac.uk

Planning a study

- Successful application Don't miss the NIHR benchmark
- Costing your research project
- Health Economics
  - DIRUM database: resource for health economics research
  - Economic Evaluation
- Study Design
  - Cross-over trials
  - EQUATOR
  - Help with methodological approaches
  - Qualitative Research Design
    - Discourse Analysis
  - Quantitative studies
    - Clinical Trials
      - Cluster randomised trials
      - Complex Interventions
      - Cross-over Trials
      - Factorial design clinical trials
      - Meta-analysis
      - Non-Inferiority Trials
      - Parallel Group Study
      - Phase I and Phase II Trials
      - Systematic Review
    - Systemic Reviews



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make it clear
Plain English Summaries in NIHR funded research
Spotlight on
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View our publications and our libraries of references
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Resource centre
News and events
Public engagement funding within research grants – new video from Wellcome Trust
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Enhancing the QUAlity and Transparency Of health Research

Visit the EQUATOR Spanish Website

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Essential resources for writing and publishing health research

Library for health research reporting

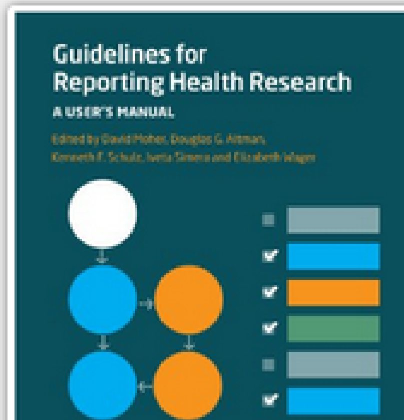
The Library contains a comprehensive searchable database of reporting guidelines and also links to other resources relevant to research reporting.

- Search for reporting guidelines
- Not sure which reporting guideline to use?
- Reporting guidelines under development
- Visit the library for more resources

Reporting guidelines for main study types

Randomised trials	CONSORT	Extensions	Other
Observational studies	STROBE	Extensions	Other
Systematic reviews	PRISMA	Extensions	Other
Case reports	CARE		Other
Qualitative research	SRQR	COREQ	Other
Diagnostic / prognostic studies	STARD	TRIPOD	Other
Quality improvement studies	SQUIRE		Other
Economic evaluations	CHEERS		Other
Animal pre-clinical studies	ARRIVE		Other
Study protocols	SPIRIT	PRISMA-P	Other

See all 285 reporting guidelines



New book edited by the EQUATOR team Guidelines for Reporting Health Research: a User's Manual

Key points

- Planning stage is crucially important
- Discuss your research ideas with a wide range of people, find good collaborators (supervisor, mentors)
- Consult with specialists early (information specialist, statistician, research support)
- Consult with patients
- Develop a robust protocol and research plan